

Time-Dependent Mechanical Evolution of Beef Using Constitutive Artificial Neural Networks

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Abstract

In everyday cooking, the tenderness and texture of beef are core indicators determining the consumer's eating experience. Shear force and the macroscopic mechanical response of the tissue are highly correlated with sensory feedback during chewing, and these mechanical properties evolve significantly with storage conditions (such as time). However, the mechanical degradation patterns of beef of different textures and anatomical cuts differ significantly under the same refrigeration conditions, making it difficult for consumers to scientifically plan the optimal cooking sequence and storage period after bulk purchases. To address this, this study proposes a constitutive artificial neural network (CANN) incorporating a time decay function to quantify the mechanical degradation behavior of beef from different anatomical cuts (such as shoulder and sirloin) during refrigeration. We use deformation gradient and storage time as network inputs, calculate the fundamental strain energy function using the partial invariants of the right Cauchy-Green tensor, and couple an exponential time penalty term to characterize the softening effect over time. This neural network directly embeds physical constraints into the model architecture, enabling flexible and accurate fitting of experimental mechanical data while ensuring strict objectivity and mechanical consistency. The model's predictions provide consumers with a scientifically tailored guide to food handling: accurately identifying easily decaying parts suitable for priority cooking, as well as durable parts that can be stored for longer periods. This research not only provides an innovative data-driven modeling framework for the mechanical behavior of biological soft tissues evolving over time, but also offers a rigorous engineering mechanics perspective for optimizing home food management and improving cooking quality.

Keywords: Constitutive Artificial Neural Network, Mechanical Degradation, Beef Tenderness

1 Motivation

Beef, being a soft tissue, is a highly heterogeneous and complex material. Its mechanical properties exhibit a significant time dependence. Tenderization begins shortly after slaughter and gradually intensifies after rigor mortis [1]. Studies have shown that the macroscopic mechanical response of beef changes significantly after aging and the end of rigor mortis; during this aging process, the shear elasticity of the samples decreases by nearly 50% [2]. Furthermore, the rate of elasticity decrease is not linear, with the rate of tissue softening typically being greater in the first few days of aging [2]. Structurally, this softening is driven by the progressive degradation of myofibrillar proteins and connective tissue, which reduces the meat's resistance to shear and compressive forces[3].

par However, there is currently no consensus on the optimal aging time for different meats because the rate and extent of tissue degradation vary greatly depending on muscle type and texture (such as the fat-to-muscle ratio) [4][5].

The significant differences and uncertainties in the mechanical degradation of different cuts of beef during refrigeration pose a challenge to daily household food management. Consumers often make bulk purchases of different cuts of beef over the weekend, making it difficult to scientifically determine which cuts (prone to degradation) should be cooked first for optimal taste and which cuts (more resistant to spoilage) can be refrigerated for longer. Therefore, developing a data-driven model capable of predicting the mechanical degradation patterns of beef from different anatomical cuts is of significant practical importance in helping consumers optimize their family meat purchasing plans, determine scientific cooking sequences, and adjust refrigeration storage times.

Objectives of this study. Brief paragraph listing the aims of this study. As an example: First we will create a model that ensures we meet certain physics principles. Second we will test our model and assumptions with a certain set of data and evaluate it in a certain way. Lastly, we will compare our novel physics-based model to an analytical model, FEA model, or out-of-the-box neural network with no physics.

2 Methods

2.1 Problem Overview

This study aims to characterize the temporal evolution of the mechanical properties of beef during refrigeration using a Constitutive Artificial Neural Network (CANN). Beef tenderization is a dynamic process involving complex biochemical changes and exhibits significant time dependence. Due to the inherent high heterogeneity of biological soft tissues, the mechanical degradation rates of meat from different anatomical locations vary significantly, making it difficult for traditional single constitutive models to provide generalized predictions. By modeling this time-dependent mechanical degradation, this study aims to provide quantitative support for home consumers to optimize their food procurement strategies and cooking sequence decisions.

2.2 Assumptions

To construct a model of the evolution of beef mechanical properties, we propose the following physical assumptions based on the continuum mechanics framework:

(i) We treat beef tissue as a continuum and assume that its volume remains constant during deformation. The incompressibility constraint is represented by the Jacobian determinant:

$$J = \det(\mathbf{F}) = 1$$

where $\mathbf{F}$ is the deformation gradient tensor.

(ii) Although beef muscle tissue exhibits transversely isotropic behavior with fiber as the axis of symmetry at a microscopic level, considering that the samples in this experiment were all standardized columnar samples along the muscle fiber direction, and the aim was to capture the mechanical degradation laws at the macroscopic scale, we simplified the assumption to an isotropic medium. Under this assumption, the strain energy density function Ψ depends only on the first two principal invariants I_1 and I_2 of the right Cauchy-Green deformation tensor $\mathbf{C} = \mathbf{F}^T \mathbf{F}$:

$$I_1 = \text{tr}(\mathbf{C}), \quad I_2 = \frac{1}{2} [(\text{tr} \mathbf{C})^2 - \text{tr}(\mathbf{C}^2)]$$

(iii) We assume that macroscopic stress can be derived from the strain energy density function with respect to the deformation gradient. Incorporating the hydrostatic pressure p introduced by the incompressible constraint, the first Piola-Kirchhoff stress $\mathbf{P}$ is expressed as:

$$\mathbf{P} = \frac{\partial \Psi}{\partial \mathbf{F}} - p \mathbf{F}^{-T}$$

(iv) We assume that the effect of refrigeration time t on mechanical properties manifests as a global scalar decay. This physical process is captured by introducing a time-dependent term into the CANN architecture to characterize the significant decrease in shear elasticity due to tissue biochemical degradation. To more accurately characterize this process, we consider the following two evolutionary equations:

a) Double Exponential Decay:

This model assumes that beef tenderization is determined by multiple biochemical processes with different rates. Its mathematical expression is:

$$\phi(t) = A_1 e^{-w_{t,1}t} + A_2 e^{-w_{t,2}t}$$

Where $w_{t,1}$ and $w_{t,2}$ represent the rate constants of the rapid and slow softening phase.

b) Stretched Exponential (Kohlrausch) Function:

Considering the high heterogeneity of beef as a biological tissue, the decay of its mechanical properties may not follow a single relaxation time. Its mathematical expression is:

$$\phi(t) = \exp[-(w_t \cdot t)^\beta]$$

Where β ($0 < \beta \leq 1$) is the stretching parameter, used to describe the diversity of the degradation process, and can better fit the differences in evolution rates between different anatomical sites than a simple exponential model.

2.3 Model Architecture

Our group used Python 3 to design and train a constitutive artificial neural network (CANN), and relied on the TensorFlow and Keras deep learning frameworks to build the network topology, using NumPy for tensor data preprocessing and computation. As shown in Figure 1, this CANN architecture contains two parallel input streams to achieve decoupled computation: a deformation stream and a time stream.

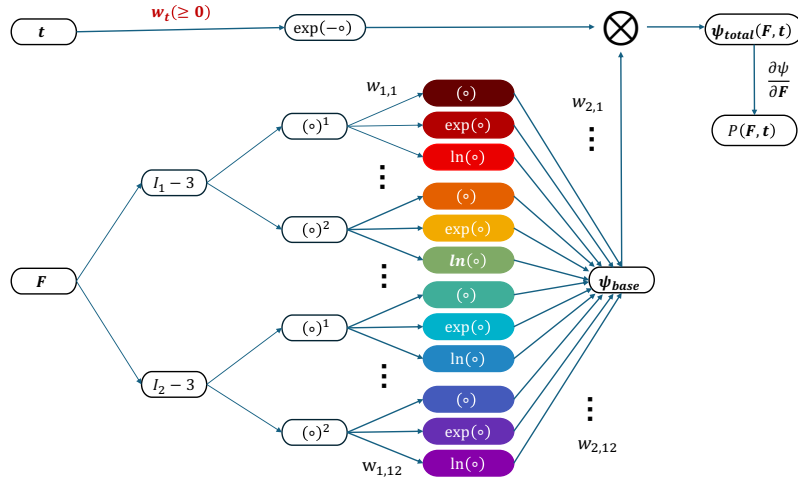


Fig. 1: Network Architecture

(i) Base Strain Energy Network:

The first and second principal invariants I_1 and I_2 of the right Cauchy-Green deformation tensor are input to the deformation stream. We constructed an isotropic, completely incompressible CANN with two hidden layers. The first layer of the network computes the exponential terms $(I_i - 3)^1$ and $(I_i - 3)^2$ of the invariant reference state; the second layer applies three activation functions in parallel: the identity function ($\circ$), the exponential function ($\exp(\circ) - 1$), and the logarithmic function ($-\ln(1 - \circ)$). The complete analytical expression for the fundamental strain energy function Ψ_{base} output by the network is:

$$\begin{aligned} \Psi_{base}(I_1, I_2) = & w_{2,1}w_{1,1}[I_1 - 3] + w_{2,2}[\exp(w_{1,2}[I_1 - 3]) - 1] - w_{2,3}\ln(1 - w_{1,3}[I_1 - 3]) \\ & + w_{2,4}w_{1,4}[I_1 - 3]^2 + w_{2,5}[\exp(w_{1,5}[I_1 - 3]^2) - 1] - w_{2,6}\ln(1 - w_{1,6}[I_1 - 3]^2) \\ & + w_{2,7}w_{1,7}[I_2 - 3] + w_{2,8}[\exp(w_{1,8}[I_2 - 3]) - 1] - w_{2,9}\ln(1 - w_{1,9}[I_2 - 3]) \\ & + w_{2,10}w_{1,10}[I_2 - 3]^2 + w_{2,11}[\exp(w_{1,11}[I_2 - 3]^2) - 1] - w_{2,12}\ln(1 - w_{1,12}[I_2 - 3]^2) \end{aligned}$$

(ii) Time Degradation Stream

The time stream receives the number of days t the samples are stored as a single input value. To accurately capture the complex evolutionary dynamics of beef during tenderization, we implement a physical information layer based on the stretching exponent (Kohlrausch) function within this stream.

The input t first passes through an unbiased fully connected layer to obtain a non-negative rate weighting term $w_t t$. Then, a custom Lambda activation layer introduces a stretching parameter β representing heterogeneity, and the time degradation factor $\phi(t)$ is calculated:

$$\phi(t) = \exp[-(w_t \cdot t)^\beta]$$

The network couples this evolution factor with the base strain energy of the deformation flow output through a Keras multiply layer, thus outputting a time-regulated total strain energy density:

$$\Psi_{total}(I_1, I_2, t) = \phi(t) \cdot \Psi_{base}(I_1, I_2)$$

(iii) Physics-Informed Output Layer

The final output layer of this model does not directly output energy, but rather maps it to macroscopic mechanical stress. In the Keras framework, we use a custom gradient function to automatically differentiate Ψ_{total} with respect to I_1 and I_2 . Combining the incompressibility condition and the free boundary assumption, taking uniaxial compression as an example, the nominal stress P_{11} output by the network is defined as:

$$P_{11} = 2 \left[\frac{\partial \Psi_{total}}{\partial I_1} + \frac{1}{\lambda} \frac{\partial \Psi_{total}}{\partial I_2} \right] \left[\lambda - \frac{1}{\lambda^2} \right]$$

2.4 Loss function

This study trains the CANN model by minimizing the residual between the predicted nominal stress and the experimentally measured stress. The loss function $L(\theta)$ consists of two parts: a mean squared error (MSE) term characterizing the model's fitting ability and an L_1 regularization term characterizing the model's simplicity. Its complete mathematical expression is defined as:

$$L(\theta) = \sum_{k \in \{TC, SS\}} \frac{\omega_k}{N_k} \sum_{i=1}^{N_k} (P_{exp,i}^{(k)} - P_{pred,i}^{(k)}(\theta))^2 + \lambda_{reg} \sum_{j=1}^M |w_j|$$

Variable Definitions:

- (i) θ : The set of all parameters to be learned in the neural network, including strain energy weights $w_{1,k}, w_{2,k}$ and time decay rate w_t .
- (ii) $P_{exp,i}^{(k)}$ and $P_{pred,i}^{(k)}$: represent the experimental stress value and model predicted stress value of the i -th data point under loading mode k (TC : uniaxial tension/compression; SS : simple shear), respectively.
- (iii) N_k : the total number of data sampling points under a specific loading mode.
- (iv) ω_k : sample weights. In the code, this value is set to the reciprocal of the absolute value of the maximum stress under each loading mode ($1/\max |P_{exp}^{(k)}|$).
- (v) λ_{reg} : penalty weight parameter, defined by the variable pen.

2.5 Data

Two beef cuts, New York strip and sirloin, were selected to investigate the relationship between freshness and mechanical response. Thirty samples were prepared in total. The experimental design consisted of three freshness conditions defined by refrigerated storage duration following thawing: freshly thawed (Day 0), one day at 4 °C, and two days at 4 °C. To generate these conditions, frozen beef stored at −18 °C was divided into three thawing batches at 24-hour intervals. Each batch contained ten samples, including five New York strip and five sirloin specimens. After cutting into standardized rectangular blocks of dimensions $1 \times 1 \times 2$ cm³, all samples were refrozen at −18 °C to standardize storage history. Prior to the testing, samples were thawed according to the designed freshness condition.

From each rectangular beef block, a cylindrical sample of 8 mm height was extracted using a biopsy punch to ensure geometric consistency for mechanical testing. A total of thirty cylindrical samples were therefore obtained. Figure 2a shows the prepared rectangular beef blocks prior to cylindrical extraction, and Figure 2b illustrates the experimental equipment.



(a) rectangular beef blocks



(b) experimental equipment

Fig. 2: Material samples and testing equipment

Uniaxial compression experiments were conducted using a Discovery HR20 rheometer equipped with parallel plates. During testing, force, displacement, and time were continuously recorded. The raw dataset for each sample consists of synchronized time-series measurements of normal force and axial displacement, from which engineering stress and strain can subsequently be calculated. An example of the raw data structure for compression test is provided in Table 1.

Table 1: Experimental Data of Beef Samples

Normal stress (Pa)	Axial force (N)	Gap (μm)	Time (s)	Strain (%)	Stress (Pa)
1364.59	0.0342958	11748.9	1.01598	-2.0925	682.293
1878.43	0.04721	11509	2.02029	-4.09166	939.213
2369.79	0.0595593	11269	3.02581	-6.09166	1184.89
2683.67	0.0674479	11029	4.0323	-8.09166	1341.83
3219.44	0.0809133	10783.7	5.03412	-10.1358	1609.72
3571.52	0.0897621	10543.6	6.03573	-12.1367	1785.76
4218.4	0.10602	10303.6	7.04083	-14.1367	2109.2
4731.92	0.118926	10063.7	8.04735	-16.1358	2365.96
5565.05	0.139865	9818.65	9.0514	-18.1779	2782.53

2.6 Analyses

To train and evaluate the proposed CANN model, the data were divided into training and testing sets using an 80%–20% split. The training set was used to learn model parameters, while the testing set was used to assess predictive performance on previously unseen samples. In addition to the baseline model, different time-degradation formulations were explored to represent the softening behavior of beef during refrigeration. Model performance was compared across these configurations to evaluate how well each formulation captures the observed mechanical degradation. Parameter studies were conducted on the time-decay function by testing different values of the decay rate and stretching parameter to determine their influence on the predicted mechanical softening behavior. The effect of regularization was also examined by varying the L1 penalty coefficient. Models with weaker and stronger regularization were compared to assess the trade-off between fitting accuracy and model simplicity. Finally, predicted stress–strain curves were reconstructed for each configuration to examine the mechanical response predicted by the model over the deformation range.

2.7 Evaluation

Model performance was evaluated by comparing predicted stresses with experimental measurements using three standard regression metrics: mean absolute error (MAE), root mean squared error (RMSE), and

the coefficient of determination (R^2). The mean absolute error (MAE) measures the average magnitude of prediction errors:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |P_{\text{exp},i} - P_{\text{pred},i}|$$

The root mean squared error (RMSE) penalizes larger errors more strongly and reflects overall prediction accuracy:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (P_{\text{exp},i} - P_{\text{pred},i})^2}$$

The coefficient of determination (R^2) indicates how well the model explains the variability of the experimental data:

$$R^2 = 1 - \frac{\sum_{i=1}^N (P_{\text{exp},i} - P_{\text{pred},i})^2}{\sum_{i=1}^N (P_{\text{exp},i} - \bar{P}_{\text{exp}})^2}$$

Higher R^2 and lower MAE and RMSE indicate better model performance. In addition to these quantitative metrics, predicted stress–strain curves were visually compared with experimental curves to assess whether the model reproduces the overall mechanical behavior of the samples.

3 Results

Here you will describe the results from your analyses. Discuss your model performance using your chosen evaluation metrics. Include relevant figures and tables.

4 Discussion

Here you will interpret your model results and comment on why you think the model behaved the way it did and what your model results mean. Do you think your model was able to capture the physical phenomena present in the data? Was one parameter set/model better than another parameter set/model?

Limitations and future work. What is the applicability/scope of your model e.g. only hyperelastic materials? Did your model fail to capture any expected behaviors? What would you do differently or add if you had more time?

Acknowledgments

Authors' contributions Briefly describe how each person contributed to the final project including writing, editing, coding, analyzing, etc.

Appendix A.

If applicable you can include an appendix.

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